

A protocol for plant and soil water relation measurements in chickpea

1. Materials

1.1 Biological samples

- Plant material: Fully expanded central leaflets from completely expanded, sun-exposed trifoliate leaves, and intact root tissues harvested at specific experimental timepoints.
- Uniformly kneaded soil core samples collected from respective treatment plots.

1.2 Equipment and instrumentation

- Wescor PSYPRO Automated Water Potential System (equipped with C52 sample chambers) or a Wescor Psychrometer/Osmometer system.
- Surgical tapes, 1.5 mL microcentrifuge tubes, high-precision analytical balance, liquid nitrogen, mortar and pestle, refrigerated microcentrifuge, and a forced-air laboratory drying oven.

2. Methods

2.1 Vapor pressure deficit

1. Monitor ambient environmental parameters (air temperature and relative humidity) daily between 11:00 AM and 1:00 PM at each sampling site.
2. Determine the vapor pressure deficit (VPD, expressed in millibars) using the psychrometric relationship:

$$VPD = e_s - e_a$$

Where e_s is the saturated vapor pressure at the recorded air temperature, and e_a is the actual ambient vapor pressure calculated from relative humidity data.

2.2 Relative water content (RWC)

1. Harvest healthy, completely expanded single leaves or intact root systems instantly at designated time intervals (e.g., during treatment, peak stress, or post-watering recovery).

2. Weigh the tissue immediately on an analytical balance to record the fresh weight (FW) (See Note 1).
3. Submerge the tissue samples in de-ionized or distilled water and incubate for 4 to 6 hours (incubate on ice for 4 hours for root tissues) to achieve complete hydration.
4. Remove the tissues, carefully blot away excess surface water using filter paper, and weigh immediately to record the maximum turgor/turgid weight (TW) (See Note 2).
5. Place the tissue samples inside a drying oven configured at 60°C to 80°C for 48 to 72 hours until the mass stabilizes. Weigh the desiccated samples to record the dry weight (DW).
6. Calculate the percentage relative water content (RWC) using the following formula:

$$\text{RWC (\%)} = \frac{\text{Fresh weight} - \text{Dry weight}}{\text{Turgid weight} - \text{Dry weight}} \times 100$$

2.3 Soil, root, and leaf water potential (Ψ) quantification

1. Collect soil cores, clean roots, or leaf tissues rapidly and place them directly into 1.5 mL microcentrifuge tubes. Seal the tube caps tightly using surgical tape and immediately freeze the samples in liquid nitrogen. Store at -80°C until processing.
2. To extract cell sap, crush the frozen root or leaf tissues inside their tubes using a pestle, then centrifuge at 12,000 rpm for 1 minute at 4°C to collect the liquid sap fraction.
3. For soil analysis, uniformly knead the collected soil core samples to ensure an even moisture distribution.
4. Pipette the extracted cell sap or place the kneaded soil matrix directly into a C52 sample chamber connected to an automated Wescor PSYPRO psychrometer system to measure total water potential (Ψ_w , expressed in bars).

5. To isolate the osmotic potential (Ψ_s), take a fraction of the extracted cell sap, freeze it completely in liquid nitrogen, and test the sample using a calibrated Wescor osmometer.
6. Calculate the mechanical turgor potential/hydrostatic pressure (Ψ_p) of the tissue using the standard water potential thermodynamic equilibrium equation:

$$\Psi_w = \Psi_s + \Psi_p$$

3. Notes

1. To prevent rapid evaporative water loss, tissue samples must be weighed immediately upon excision for RWC, or flash-frozen instantly in liquid nitrogen for water potential assays. Any delay will artificially lower fresh weights and drop baseline tissue potentials.
2. When blotting tissues after the hydration step to record turgid weight, ensure no moisture remains on the surface. Conversely, avoid pressing excessively hard on leaf borders, which can express symplastic water and lower the final calculated RWC percentage.

Please cite the original articles for this method:

- Kumar M, N. Jane W, N. Verslues P, E. (2012). Role of the putative osmosensor *Arabidopsis* histidine Kinase1 in dehydration avoidance and low-water-potential response. *Plant Physiol.* 161:942–953. 10.1104/pp.112.209791
- Chilakala AR, Mali KV, Irulappan V, Patil BS, Pandey P, Rangappa K, Ramegowda V, Kumar MN, Puli COR, Mohan-Raju B and Senthil-Kumar M (2022) Combined Drought and Heat Stress Influences the Root Water Relation and Determine the Dry Root Rot Disease Development Under Field Conditions: A Study Using Contrasting Chickpea Genotypes. *Front. Plant Sci.* 13:890551. doi: 10.3389/fpls.2022.890551